

AI-Augmented Theory Formation: Methodology, Control Mechanisms, and Epistemic Audits in the Methane Metauniverse (MMU) Project

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Abstract

This article reports on a controlled experiment in AI-human collaboration applied to the development of a speculative physical framework, the Methane Metauniverse (MMU). The objective is not to validate the physical correctness of MMU, but to investigate whether large language models, when properly constrained, can augment human reasoning in early-stage theory formation.

To this end, a strict knowledge-control architecture was implemented, including a canonical reference document, a retrieval-augmented generation (RAG) system, and a formally defined instruction set designed to suppress conceptual drift and hallucination. Independent epistemic audits were then conducted using two external AI systems (Gemini and Copilot) following a fixed multi-phase protocol.

We analyze the methodology, audit outcomes, convergence and divergence between AI assessors, and derive general lessons for AI-assisted scientific research. The MMU framework serves here as a stress-test case for AI-augmented theory development rather than as the primary scientific claim.

1 Introduction

Large language models are increasingly used in scientific contexts, from literature review to symbolic reasoning and code generation. This raises a fundamental methodological question: can AI systems meaningfully assist in the development of new scientific theories without introducing uncontrolled drift, hallucination, or false epistemic authority?

This article addresses that question through a controlled case study. The Methane Metauniverse (MMU) project was deliberately chosen as a challenging testbed. MMU is a speculative, non-standard physical framework combining discrete geometry, elastic dynamics, and projection principles. Such a framework is highly susceptible to misinterpretation, uncontrolled extension, or category errors when handled by unconstrained AI systems.

The aim of this work is therefore not to argue for the correctness of MMU as a physical theory. Instead, the focus lies on the *process*: how AI augmentation was structured, constrained, audited, and evaluated, and what epistemic value such an approach can provide during early-stage theoretical research.

2 The MMU Project as a Testbed

The MMU framework proposes a discrete geometric lattice with internal degrees of freedom as a candidate substrate for physical phenomena. Its mathematical complexity, incomplete empirical status, and departure from standard field-theoretic formulations make it an ideal stress test for AI-assisted reasoning.

Any uncontrolled hallucination, silent redefinition of terms, or unjustified extrapolation would quickly render the framework inconsistent. MMU therefore functions here not primarily as a physical hypothesis, but as a diagnostic environment for testing whether AI systems can maintain coherence under epistemic pressure.

3 Human–AI Collaboration Model

3.1 Division of Roles

A strict separation of epistemic roles was enforced throughout the project. Human contributors retained responsibility for hypothesis generation, physical intuition, interpretative decisions, and final acceptance or rejection of claims. AI systems were restricted to formal reconstruction, consistency checking, structured criticism, and audit simulation.

At no point was AI treated as an epistemic authority or a source of truth. Instead, it functioned as a systematic adversarial and organizational tool.

3.2 Iterative Development Loop

Theory development followed an explicit iterative loop:

Human proposal $\rightarrow$ AI analysis $\rightarrow$ Human revision $\rightarrow$ Re-audit.

This loop allowed weaknesses to be exposed early and prevented the accumulation of mutually inconsistent assumptions. Each iteration was documented and anchored to the same controlled knowledge base.

4 Knowledge Control Architecture

4.1 Retrieval–Augmented Generation (RAG)

All relevant MMU documents were stored in a retrieval system accessible to the AI. This ensured a fixed, inspectable knowledge base and avoided reliance on undocumented latent training data. AI outputs were thus traceable to explicit sources rather than implicit priors.

4.2 Canonical Reference (Canon)

A canonical document was defined as the immutable foundation of the project. It explicitly lists assumptions, non-assumptions, definitions, projection rules, and scaling laws. Any deviation from the Canon had to be explicitly identified and justified.

The Canon served as a hard constraint preventing silent conceptual drift.

4.3 Instruction Set

A formal instruction set constrained AI behavior. It enforced physical consistency, explicit uncertainty labeling, dimensional analysis, and a strict separation between ontology and epistemology. This instruction layer proved essential in suppressing metaphor escalation and unintended conceptual blending.

5 Drift and Hallucination Control

Typical AI failure modes include silent redefinition of terms, injection of foreign conceptual layers, and metaphor amplification. These were countered through mandatory Canon referencing, structured audit protocols, and explicit status markers (e.g. *RESOLVED*, *PARTIAL*, *OPEN*) for each claim.

This control architecture significantly reduced hallucination without suppressing critical analysis.

6 Documentation Strategy, Publication Record, and Project Timeline

A central component of the MMU development process was the systematic documentation of all intermediate and final results in publicly accessible repositories. The project deliberately adopted an *open documentation* strategy to ensure transparency, reproducibility, and independent scrutiny by both human researchers and AI systems.

6.1 Open Scientific Documentation

All core results of the MMU project were documented as individual, citable preprints hosted on the Open Science Framework (OSF) and mirrored on ResearchGate. Each document was written in standard scientific format and assigned a persistent DOI, allowing unambiguous referencing and long-term availability.

The publications cover distinct conceptual and technical aspects of the framework, including geometric foundations, dynamical equations, projection principles, spectral predictions, and consistency relations. This modular publication strategy served two purposes: first, it enabled fine-grained version control of theoretical developments; second, it allowed AI systems to retrieve well-scoped, self-contained units of knowledge rather than implicitly reconstructing the theory from informal sources.

A non-exhaustive overview of the documented MMU articles is provided in the bibliography, spanning works on geometric lattice structure [1, 5], atomic spectra and resonances [2, 3, 15, 23], field equations and dissipation [6, 31, 32], spin, torsion, and fermionic structure [13, 14, 17, 29], as well as gravitational and cosmological implications [19, 20, 27, 28].

6.2 Role of Documentation for AI–Augmented Research

The explicit documentation on OSF and ResearchGate played a crucial role in enabling controlled AI augmentation. By serving as the sole authoritative knowledge base for retrieval–augmented generation (RAG), the document corpus reduced the risk of hallucination, uncontrolled extrapolation, and hidden training–data bias. In effect, the written record functioned as an external, machine–readable memory that constrained and stabilized AI reasoning.

This approach highlights an important methodological insight: for AI systems to contribute meaningfully to scientific theory development, the underlying knowledge must itself be structured, explicit, and citable. Informal or implicit knowledge representations are insufficient for reliable AI interaction.

6.3 Project Timeline

The entire MMU development and audit process was conducted within a clearly bounded time frame. From the initial consolidation of the geometric core to the completion of independent AI audits, the total project duration amounted to approximately **three months**.

Within this period, the following milestones were achieved:

- formulation and consolidation of the MMU canonical assumptions,
- generation and publication of a structured series of OSF preprints,
- construction of a retrieval–augmented documentation corpus,
- development of a strict AI instruction and control framework,
- execution of independent epistemic audits using multiple AI systems.

The relatively short development cycle, combined with extensive documentation and external auditing, illustrates both the potential efficiency and the limitations of AI–augmented theoretical research. While AI systems can accelerate consistency checks and structural analysis, the generation of physically meaningful hypotheses and interpretations remained a human task.

7 AI–Based Epistemic Audits

7.1 Audit Objectives

The epistemic audits were designed to answer explicit and testable research questions concerning AI–assisted scientific reasoning under controlled information constraints. The central objectives were:

- to determine whether an independent AI system can reconstruct and understand a non–standard physical framework solely from publicly available scientific documentation hosted on OSF and ResearchGate,
- to test whether such a system can formulate critical, non–trivial scientific objections without access to internal development context,

- to evaluate whether coherent and self-contained responses can be generated exclusively from the documented material,
- to assess whether a second, independent AI system can in turn audit and evaluate the first AI’s reconstruction and critique,
- and to measure whether conceptual drift and hallucination are effectively suppressed when AI reasoning is constrained by a fixed canonical reference and retrieval-augmented documentation.

A further objective was to examine whether AI systems remain anchored to established physical principles and experimentally verified facts, rather than introducing speculative extensions or category errors when analyzing an incomplete theoretical framework.

7.2 Audit Protocol

To address these objectives, a multi-phase audit protocol was defined in advance and frozen prior to execution. The protocol consists of the following stages: prior questions (ontology and scope), blind reconstruction from public documentation, systematic criticism, structured response evaluation, and post-response re-assessment.

Each phase targets specific epistemic properties, including conceptual clarity, internal consistency, mathematical rigor, empirical anchoring, and falsifiability. The protocol was applied identically across all audit systems to ensure comparability and methodological consistency.

7.3 Internal Baseline Audit

Before conducting external audits, an internal epistemic audit was performed with full access to the MMU Canon, the complete documentation corpus, and the development context. The purpose of this audit was not validation, but structural consolidation: identifying internal inconsistencies, clarifying definitions, and establishing a best-case baseline for conceptual coherence under ideal information conditions.

This internal audit is epistemically distinct from the external audits. It represents an upper-bound reference case with maximal contextual access and cannot be considered independent. Its results are therefore used solely as a comparative baseline rather than as confirmation.

7.4 Independent External Auditors

Two external AI systems, Gemini and Copilot, were selected as independent auditors. Both systems were restricted to publicly available sources, specifically the OSF and ResearchGate MMU documentation. They had no access to internal development discussions, instruction tuning, or to each other’s outputs.

This ensured that each audit constituted an independent epistemic reconstruction and critique based purely on documented material.

7.5 Comparative Results and Scoring

Each audit produced a structured evaluation across six epistemic criteria, scored on a scale from 0 (absent) to 5 (fully satisfied). Table 1 summarizes the comparative results of the internal baseline audit and the two external audits.

Table 1: Comparative Epistemic Audit Scores (0–5)

Criterion	Internal Audit	Gemini	Copilot
Conceptual Clarity	4.5	4.5	4.0
Internal Consistency	4.0	4.0	3.0
Mathematical Rigor	3.5	3.0	2.0
Empirical Anchoring	3.0	3.5	1.0
Falsifiability	4.0	5.0	2.0
Novelty	5.0	5.0	3.0

7.6 Analysis of Convergence and Divergence

Despite differing architectures and evaluation styles, the external audits show strong convergence with the internal baseline on the identification of core unresolved issues, most notably the absence of appendix-grade derivations, explicit operator constructions, and numerical predictions with uncertainty propagation.

Divergences between Gemini and Copilot primarily reflect differences in audit philosophy: Gemini emphasizes conceptual coherence and falsifiability, whereas Copilot applies stricter standards to formal mathematical completeness and empirical readiness. Importantly, no external audit introduced spurious concepts or undocumented assumptions, indicating successful suppression of hallucination and uncontrolled drift.

7.7 Independent Qualitative Classification by AI Systems

Beyond quantitative scoring, both external AI systems provided an explicit qualitative classification of the MMU framework based on their independent reconstruction and critique.

Gemini Assessment. Gemini classified MMU as a *speculative but internally coherent theoretical framework*. It emphasized the logical consistency of the geometric and structural assumptions, the explicit separation between foundational postulates and derived claims, and the presence of clearly identifiable falsification points. Gemini noted that MMU does not currently meet the standards of a predictive physical theory due to missing quantitative results, but explicitly framed it as a viable candidate for further formal development rather than as a category error or pseudoscientific construct.

Copilot Assessment. Copilot independently classified MMU as a *conceptually promising but formally incomplete framework*. Its evaluation placed stronger emphasis on the absence of explicit operators, boundary conditions, and numerical predictions. Copilot explicitly stated that MMU remains outside the domain of established physical theories at its current stage, but nevertheless

qualifies as a legitimate speculative research program provided that the identified formal gaps are addressed.

Comparative Interpretation. Despite differences in emphasis, both AI systems converged on a remarkably similar high-level classification. Neither system rejected MMU as incoherent, unphysical, or metaphysical in nature. Instead, both positioned it in an intermediate epistemic category: a structured speculative framework that is methodologically admissible, logically constrained, and in principle falsifiable, yet not mature enough to be considered a complete physical theory.

This convergence is notable given the independence of the audits and the absence of shared context. It suggests that, under controlled documentation and canonical constraints, independent AI systems are capable of producing stable and comparable epistemic classifications of complex theoretical proposals.

7.8 Auto-Evaluated Classification (Constrained Internal Assessment)

In addition to the independent external audits, a constrained internal auto-evaluation was performed by the AI system used during MMU development. This assessment was conducted under strict adherence to the same canonical reference, instruction set, and documentation constraints described throughout this work.

The purpose of this auto-evaluation was not confirmation, but reflective consistency checking: assessing whether an AI system with full access to the Canon and the structured RAG corpus would arrive at a materially different epistemic classification than independent web-only auditors.

Auto-Evaluation Result. Under these constraints, the MMU framework was classified as a *structurally coherent but formally incomplete speculative physical theory*. The evaluation identified the same primary limitations as the external audits, most notably:

- missing appendix-grade derivations for key operators and spectra,
- absence of fully worked numerical predictions with uncertainty propagation,
- incomplete experimental protocols for decisive falsification.

At the same time, the auto-evaluation confirmed that MMU does not exhibit category errors, hidden metaphysical assumptions, or violations of established physical principles at the level of its stated claims. All central hypotheses were found to be explicitly marked as speculative and, in principle, falsifiable.

Comparative Consistency. Importantly, the auto-evaluated classification did not exceed or contradict the qualitative assessments produced by Gemini and Copilot. Instead, it converged on the same intermediate epistemic category: neither rejection as incoherent nor acceptance as a complete physical theory.

This convergence across internal and external AI evaluations, despite differences in information access and audit role, suggests that the applied knowledge-control architecture effectively

stabilizes AI reasoning and suppresses confirmation bias, hallucination, and uncontrolled theoretical drift.

7.9 Implications for AI-Augmented Scientific Research

The layered audit structure presented in this work—combining an internal baseline audit with independent, web-only external AI audits—provides empirical insight into the capabilities and limitations of AI-augmented theoretical research.

First, the results demonstrate that independent AI systems are capable of reconstructing and critically evaluating a complex, non-standard theoretical framework using only publicly available, well-structured documentation. This indicates that AI systems can act as effective epistemic auditors when their reasoning is constrained by explicit sources and canonical references.

Second, the strong convergence between internal and external audits suggests that the applied knowledge-control architecture successfully suppresses hallucination, uncontrolled conceptual drift, and implicit extrapolation. No audit introduced spurious concepts, hidden assumptions, or violations of established physical principles beyond those explicitly stated in the documentation.

Third, the observed divergences between AI systems reflect differences in evaluation philosophy rather than inconsistency of understanding. This highlights a potential role for multi-AI audit ensembles, where convergence and divergence patterns themselves become informative epistemic signals.

Finally, the results underscore a clear boundary: while AI systems can significantly enhance structural coherence, completeness checking, and critical scrutiny, they do not replace human physical intuition, hypothesis generation, or responsibility for truth claims. AI functions here as an epistemic amplifier, not as an autonomous scientific agent.

These findings suggest that AI-augmented audits may serve as a valuable complement to traditional peer review, particularly in early-stage, speculative, or interdisciplinary research programs where formal consistency and documentation quality are critical.

8 Conclusion and Outlook

This work has presented a controlled case study of AI-augmented theory formation using the Methane Metauniverse (MMU) project as a stress test. The central contribution is methodological rather than physical: it demonstrates how large language models, when embedded within a rigorous control framework, can contribute meaningfully to early-stage scientific research without introducing uncontrolled drift or false epistemic authority.

By combining explicit canonical constraints, retrieval-augmented documentation, a formal instruction set, and independent multi-AI audits, the project achieved a level of transparency, reproducibility, and critical self-assessment that would be difficult to maintain through informal human reasoning alone.

The MMU framework itself remains a speculative and formally incomplete research program. The audits neither validate nor falsify its physical claims. Instead, they provide a structured assessment of epistemic maturity and identify concrete requirements for further development,

including explicit operator constructions, quantitative predictions, and experimentally viable falsification protocols.

Looking forward, the methodology presented here is not specific to MMU. It may be applied to other speculative or interdisciplinary theories, as well as to the development of complex models in fields beyond physics. Future research should explore standardized AI-based audit protocols, comparative studies across different AI architectures, and the integration of such audits into open scientific workflows as a complement to traditional peer review.

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A AI-Based Independent Audit Protocol

This appendix documents the fixed audit protocol used for all AI-based epistemic evaluations reported in this work. The protocol was frozen prior to the execution of any external audits and applied identically to all AI systems to ensure methodological consistency and reproducibility.

The full protocol text is archived as a standalone document and is summarized here for completeness and transparency [?].

A.1 Purpose and Scope

The purpose of the protocol is to evaluate whether an untrained, independent AI system can:

- reconstruct a non-standard theoretical framework using only public scientific documentation,
- identify substantive scientific weaknesses and open problems,
- reassess its scientific judgment after structured responses,
- and produce an independent final epistemic classification.

The protocol is explicitly *not* a validation procedure. It does not aim to establish physical correctness, but to assess conceptual reconstructability, internal consistency, responsiveness to criticism, and epistemic maturity.

A.2 General Constraints

All auditing AI systems were required to adhere to the following mandatory constraints:

- No use of private training data, internal memory, or prior conversations.
- No access to MMU-specific internal guidance, Canon documents, or RAG systems.
- Exclusive use of publicly available sources, limited to OSF preprints, ResearchGate uploads, and publicly indexed mirrors.
- Use of formal scientific language with explicit distinction between assumptions, derived results, interpretations, and open problems.

A.3 Phase 0: Prior Questions

Phase 0 tests baseline comprehension prior to any reconstruction or criticism. The AI answers a fixed set of ten prior questions covering ontology, dimensionality, observables, projection principles, dynamics, constants, relation to established physics, quantization, predictive scope, and claimed theoretical status.

These answers are not evaluative. They are later classified as CORRECT, PARTIALLY CORRECT, INCORRECT, or MISSING to assess reconstructability.

A.4 Phase I: Blind Reconstruction

In Phase I, the AI reconstructs the theory using public sources only. No criticism is permitted. The reconstruction must include:

- core assumptions and axioms,
- defined objects and structures,
- mathematical framework and scaling laws,
- claimed relation to established physics,
- stated predictions,
- and explicitly identified ambiguities.

A.5 Phase II: Critical Objection Generation

Phase II requires the generation of exactly ten scientific criticisms. Each criticism must:

- target a specific theoretical aspect,
- be potentially falsifying,
- be categorized as Conceptual (C), Mathematical (M), Physical (P), or Empirical (E),
- include an explanation of its seriousness,
- and specify what type of response would resolve it.

Non-scientific or rhetorical criticism is explicitly disallowed.

A.6 Phase IIb: Structured Author Responses

In Phase IIb, structured responses to each criticism are provided. Each response must:

- reference internal theoretical logic (axioms, projection principles, scaling laws),
- relate the claim to established physics (limits, EFT correspondence, projection),
- outline a concrete calculation or experimental test,
- and assign a resolution status: RESOLVED, PARTIAL, or OPEN.

Unresolved issues must not be rhetorically dismissed.

A.7 Phase III: Post–Response Re–Evaluation

After reviewing the responses, the AI reassesses the theory and re–scores six epistemic criteria on a 0–5 scale:

1. conceptual clarity,
2. internal consistency,
3. mathematical rigor,
4. empirical anchoring,
5. falsifiability,
6. novelty.

Score changes relative to the initial assessment must be reported. Each criticism is reclassified as RESOLVED, PARTIALLY RESOLVED, or UNRESOLVED.

A.8 Phase IV: Independent Scientific Verdict

In the final phase, the AI provides an independent verdict addressing:

- whether the framework qualifies as a scientific theory,
- its most appropriate epistemic classification,
- the strongest argument in favor of the theory,
- the strongest argument against it,
- and the single most decisive experiment or calculation.

This verdict must be written independently, without consultation with the theory authors.

A.9 Reporting Standard

The final audit report must be suitable for publication, avoid promotional language, and explicitly state limitations and uncertainties. The protocol is designed to be transferable to other speculative or interdisciplinary research programs.